

Dual-Channel Adder Verification Environment

Kratos

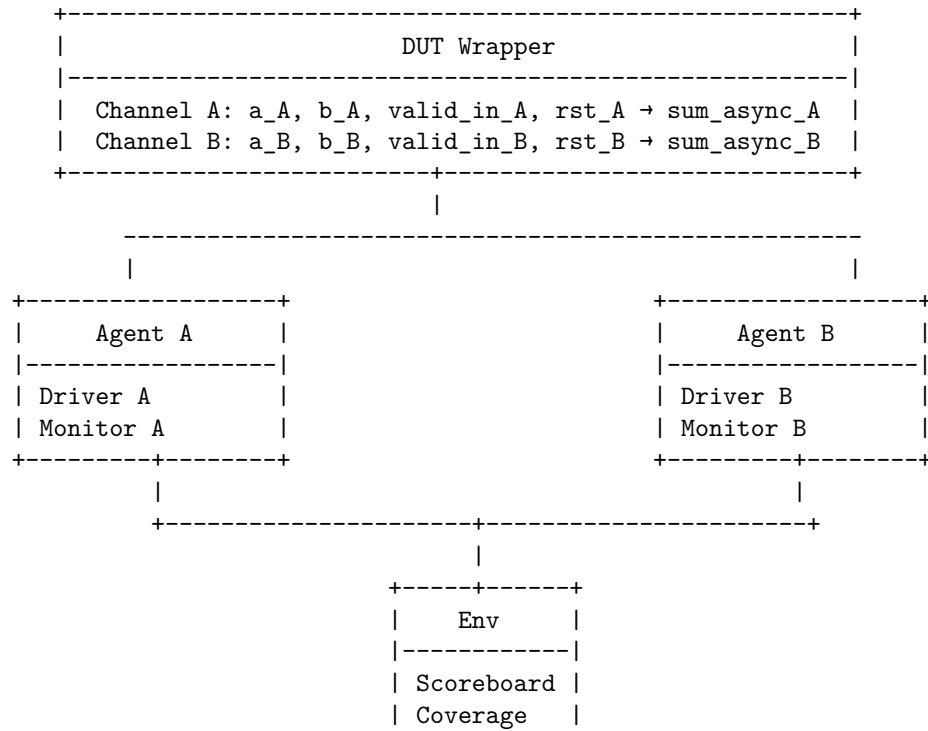
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Introduction

This document describes a complete verification environment for a dual-channel adder design using SystemVerilog, Cocotb, and Verilator. It includes architecture, coverage, regression methodology, and waveform generation.

Architecture Overview

Block Diagram



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Coverage Model

Coverage includes operand bins, pattern bins, overflow bins, illegal bins, temporal bins, and cross coverage. Threshold gating requires 75% minimum coverage.

Regression

Regression is executed using:

```
./tools/run_regression.sh
```

Waveform Generation

Waveforms are enabled using:

```
export VERILATOR_TRACE=1
```

Waveforms appear in:

```
sim_build/dump.vcd
```